UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

**UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**

SATURDAY, 14 OCTOBER 2023 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

**Instruction to Candidates :**

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

**Section A (Compulsory Questions)**

Q1. (a) Discuss **TWO (2)** activities of business requirement strategy used to analyze the user requirement in the business process reengineering. (4 marks)

(b) Identify **FIVE (5)** major elements used in creating a system request documentation during the project initiation at the planning phase of the system development life cycle (SDLC). (5 marks)

(c) Explain the differences between the phased development and throwaway prototyping in term of the system methodology. (4 marks)

(d) Discuss **THREE (3)** types of architectural views used in the architecture centric approach in terms of the object-oriented approach. (6 marks)

(e) Explain a situation where joint application development (JAD) is a suitable requirement determination technique used to collect the information compared to document analysis. (2 marks)

(f) Suggest **TWO (2)** ways to prevent scope creep from happening in a system development project. (4 marks)

[Total : 25 marks]

Q2. (a) Bank users own three types of bank accounts: an e-wallet account, a savings account, and a fixed deposit account. A bank account record contains an account number, holder name, and others. The balance of the amount for an e-wallet account, a savings account, and a fixed deposit account have interest-free, monthly interest, and interest depending on the maturity date, respectively. In addition, each type of account has transaction records such as type of transaction, amount, and others. Given the mentioned scenario, use the class diagram with invariants notation to illustrate the data storage structure of the system with the items listed below.

(i) Association relationships between two classes. (3 marks)

(ii) Generalization relationships between two classes. (4 marks)

(iii) Aggregation relationships between two classes. (2 marks)

(iv) Visibility and invariants. (2 marks)
[Hints: Combine the answer (i), (ii), (iii) and (iv) into a class diagram with invariants]

(b) Discuss the differences the encapsulation and information hiding in term object-oriented approach. (4 marks)

(c) Discuss the **FOUR (4)** phases in the rational unified process and provide the major deliverables of the phase discussed. (8 marks)

(d) Propose **TWO (2)** types of metrics used to measure the quality of the design of class and methods. (2 marks)
[Total : 25 marks]

Q3. (a) KTMB Mobile is an application that allows customers to book all train tickets online, including KTM train, ETS train, Intercity & North Komuter between Kuala Lumpur right up to Padang Besar and other destinations within the operation in peninsular Malaysia. First, the customer is required to login to his/her KTMB Mobile user account when he/she wants to book a train ticket. After that, the customer can search for the availability of the train based on the travel date, source and destination to book a train ticket. Then, the customer selects the train with an appropriate seat. The system will generate the fare price at the end of the train ticket booking. If the customer is a child (age under 12 years old) or a senior citizen (age above 60 years old), a 50% discount on the fare price will be given to the customer. After the customer makes the payment, the system will generate a copy of a ticket for the customer. Given the mentioned scenario, use the use case diagram notation to draw the system functionalities with items listed below.

(i) Actor(s) who interact with the system. (1 mark)

(ii) Use case that associated with actor(s) using association relationship(s). (5 marks)

(iii) Use case that associated with others use case(s) using either include or extend relationship(s). (6 marks)

(iv) System boundary. (1 mark)

[Hints: Combine the answer (i), (ii), (iii), and (iv) into a use case diagram]

(b) Given the flow of events,

I. The consumer scans the barcode of the item at the payment machine.
II. The system checks validity of the barcode.
III. The system notifies the consumer with a beep sound and adds the item to the cart if the barcode is valid. Otherwise, the system notifies the consumer with three beep sounds at the machine.
IV. The system computes a sub-total of the payment amount.
V. The consumer repeats step I to step IV until there are no more items to checkout.
VI. The consumer makes the payment using a credit card or debit card.
VII. The system generates a copy of receipt for the consumer.

Based on the flow of events given, use the activity diagram notation to draw the self-service checkouts process at in-store market with items listed below.

(i) Swimlane. (2 marks)

(ii) Step-by-step activities of the champion recruitment process. (4 marks)

(iii) Decision nodes used for the champion recruitment process. (4 marks)

(iv) Merge node, initial node and final-activity node usage. (2 marks)

[Hints: Combine the answer (i), (ii), (iii) and (iv) into an activity diagram]

[Total : 25 marks]

**Section B (Choose any ONE Question)**

Q4. (a) Given the flow of events,

I. Customer checkouts a shopping cart.
II. System retrieves the purchased product reward points.
III. System checks the product reward type. The product reward points will be multiplied by 1, 2 and 5 when the product reward type is standard, special and promotional, respectively.
IV. System adds the product reward points to the customer's membership account.
V. Repeat step II to step IV until there is no next product to compute reward points.
VI. System notifies the customer of the new updated reward points.

Based on the flow of events given, use the activity diagram notation to draw the reward point gaining process with items listed below.

(i) Actor(s), system, object(s). (4 marks)

(ii) Message passing and guard condition. (5 marks)

(iii) Loop fragment, lifeline and execution occurrence usage. (3 marks)
[Hints: Combine the answer (i), (ii) and (iii) into a sequence diagram]

(b) Discuss the first **FOUR (4)** levels of normalization in database design.
(8 marks)

(c) Identify **FIVE (5)** factors used to select a design strategy (e.g.: customer development). (5 marks)
[Total : 25 marks]

Q5. (a) Explain the differences among the aesthetics, consistency and minimal user efforts in terms of the principles of the user interface design. (6 marks)

(b) Identify **THREE (3)** hardware architectural components and **FOUR (4)** software architectural components used for physical architecture design.
(7 marks)

(c) Discuss **THREE (3)** factors used to select a physical architecture design (e.g.: server-based architecture). (6 marks)

(d) Discuss **THREE (3)** tasks that will be conducted by the project manager when coding the actual system. (6 marks)
[Total : 25 marks]